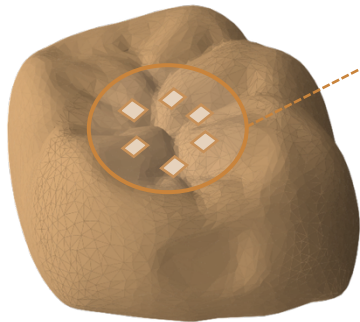
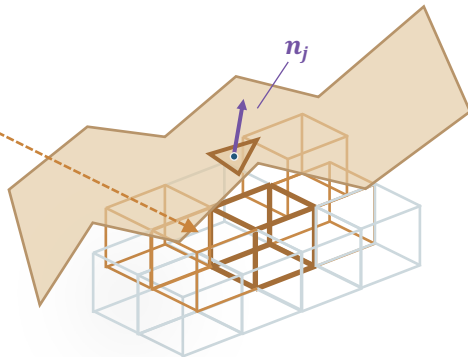


(a) Sparse surface support



Surface-neighborhood activate set

(b) Local oriented signed FDG



inactive



active

(c) Geometry code and sparse latent



g_j

Sparse Geometric VAE

$Z = (O, F)$ sparse latent

sparse support O



local feature F



geometry, local sign, and orientation are compressed jointly